EquiShare

Institutional Equipment & Space Sharing Governance Platform

CodeHack Hackathon (48-Hr Round)
Track: *Unlock Underused Equipment & Spaces*
Lead: **Muskan Gupta**
Repo: github.com/muskan5572/codehack-equishare
Live: codehack-equishare.vercel.app

1. Executive Summary & Problem Statement

Universities, research laboratories, makerspaces, and community fabrication hubs own millions of dollars in advanced capital assets—such as **5-axis CNC milling centers, scanning electron microscopes (SEM), 120W precision laser cutters, and 8K cinema camera rigs**—that remain idle **60% to 75% of the time**.

At the same time, student researchers, interdisciplinary creators, and early-stage startup teams cannot access these tools because commercial rentals or capital purchasing are prohibitively expensive. Standard booking calendars fail completely in institutional environments because they ignore 4 core requirements: **(1) Safety & Operator Qualification Liability**, **(2) Before/After Condition Baseline Audits**, **(3) Anti-Monopolization Quotas**, and **(4) Human-in-the-Loop Governance**.

2. The EquiShare 10-Step Governance Lifecycle

EquiShare closes this governance gap by transforming ordinary calendar booking into an automated institutional trust loop:

Step	Stage	Institutional Governance Function
01	Resource Listing	Owners define capacity, location, safety credentials, max duration limits, and refundable deposits.
02	Eligibility Check	Automated verification of safety certificates against institutional database before booking.
03	Access Request	Applicant selects timeslot, states academic research purpose, and nominates supervising faculty.
04	Fair-Use Validation	System validates active reservation limits (max 2) and calculates FairShare reputation quota impact.
05	Manager Review	Human-in-the-loop review desk for facility directors with mandatory explainable rationale input.
06	Fair Allocation	Approved, rejected with documented rationale, or placed on priority queue (auto-promotes on cancel).
07	Check-In Baseline	Pre-flight safety inspection: emergency stop test, calibration verify, 1-5 star cleanliness & photo.
08	Return Inspection	Side-by-side Before vs. After condition comparison card with automatic degradation detection.
09	Incident Governance	Damage or delay tickets automatically trigger investigation flags, deposit holds, and score adjustments.
10	Audit Trail Ledger	Tamper-evident, cryptographically timestamped compliance history with expandable raw JSON metadata.

3. FairShare™ Anti-Monopolization Algorithm

To prevent well-funded senior labs from monopolizing peak machine hours, EquiShare implements an explainable mathematical quota engine (fairUse.ts):

```
FairShare Score = Base (100) - Active Hold - Tool Weight + Cross-Dept Bonus (+10) + Return Reward (+2)
```

- **Mandatory Cooldowns (12h–48h):** Restricts repeated consecutive bookings of the same heavy tool to give others a turn.
- **Cross-Silo Incentive (+10 Pts):** Rewards interdisciplinary collaboration (e.g. Bio students utilizing Robotics equipment).
- **Concurrent Cap (Max 2):** Restricts a single team from reserving more than 2 active high-value assets simultaneously.

4. 1-Click Judge Sandbox & Demonstration Personas

To enable judges to evaluate the multi-party institutional trust workflow without tedious credential logins, EquiShare includes a persistent **1-Click Role Switcher** at the top of every screen:

Persona	Role	Department	Evaluation Function & Test Scenario
Muskan Gupta	Lab Director & Approver	Robotics & Fabrication Institute	Evaluates applicant FairShare scores, inspects submitted qualifications, types explainable decision notes, approves bookings (with confetti feedback), and handles damage disputes.
Ishan Roshan	Senior Researcher	Materials Science & Bioengineering	Holds verified *Laser Safety Level 2* & *Cleanroom Level 100* certificates. Demonstrates immediate booking eligibility, performs pre-use check-in baseline, and completes return inspection (+2 score reward).
Shray Shayus	Student / Creator	Mechanical Engineering	Unverified operator without required certifications. Demonstrates eligibility blocking ('Missing Certification'), submitting safety training proofs, and transparent waitlist queuing.

5. Recommended 3-Minute Judge Evaluation Walkthrough

• **1. Explore as Ishan Roshan:** Select Ishan. Notice the green *'Verified & Ready'* pill on the Scanning Electron Microscope. Click *'Request Access'*, enter a research purpose, and submit.

• **2. Switch to Muskan Gupta (Approver):** Click Muskan. Go to the *'Approvals'* tab. Review Ishan's 94/100 FairShare score and qualifications. Type an explainable rationale and click *'Approve Reservation'*.

• **3. Switch to Ishan Roshan:** Open *'Reservations'*. Click *'Step 7: Check-In Baseline'*. Complete the safety checklist (emergency stop, calibration), set cleanliness to 5★, and confirm to unlock.

• **4. Complete Return Handover:** Click *'Step 8: Return Inspection'*. View the side-by-side Before vs. After condition comparison card. Sign off on return to trigger deposit release and earn +2 reputation.

• **5. Verify Audit Trail:** Open the *'Audit Trail'* tab to view the tamper-evident, cryptographically timestamped ledger recording every actor, decision rationale, and state change.

6. Architecture, Tech Stack & Hackathon Disclosures

• **Frontend Architecture:** Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, Lucide Icons, Canvas Confetti.
• **Design System:** Deep Industrial Amber & Haas Gold aesthetic inspired by aerospace and advanced robotics foundry workshops.
• **Backend & Data Layer:** Supabase (PostgreSQL with Row-Level Security), hybrid store with instant hydration and cloud sync.
• **Safety & AI Compliance (Rule 12):** Important access and penalty decisions require human-in-the-loop explainability rather than unexplained automated systems. Licensed under the MIT Open Source License.
• **Continuous Development History:** Built authentic codebase with 23 commits pushed to GitHub during the official 48-hour build window.

Official Public Links (Verified 24/7 Accessible):
• **Live Deployed Platform:** https://codehack-equishare.vercel.app/
• **Public GitHub Repository:** https://github.com/muskan5572/codehack-equishare